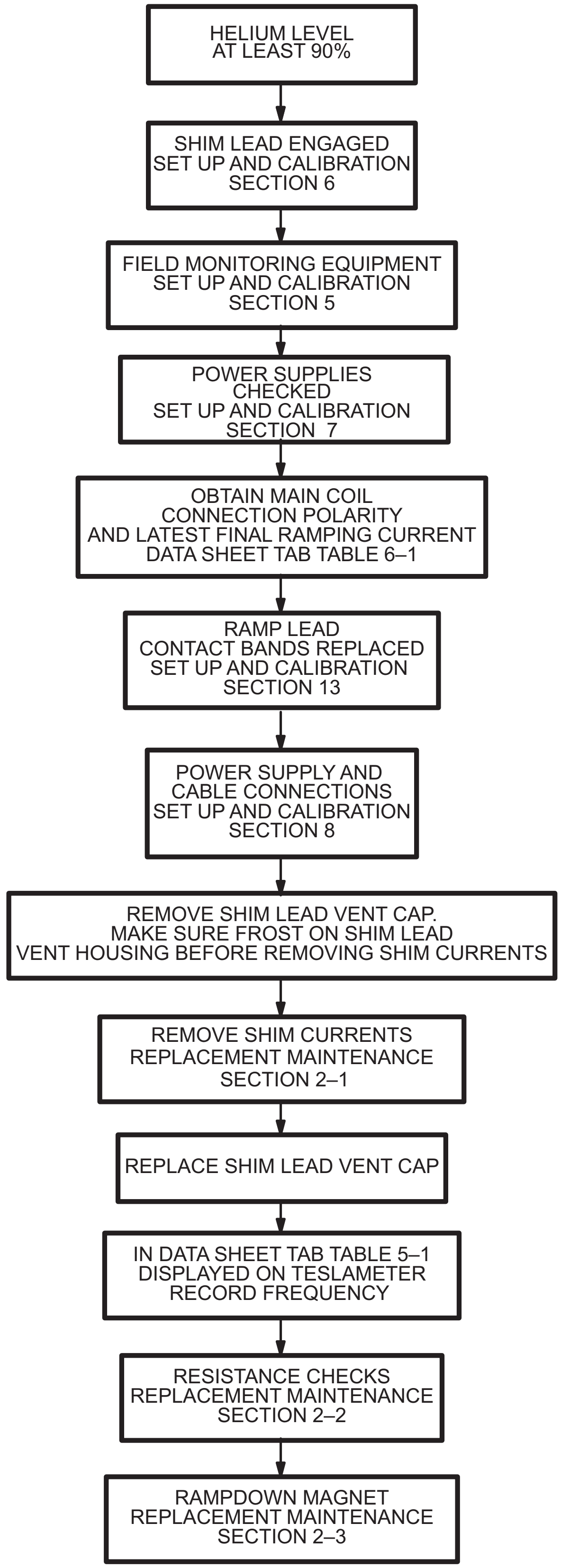
SECTION 2 – MAGNET RAMPDOWN (DECREASE TO ZERO)

WARNING!

THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RAMPING DOWN THE MAGNET:

1. **THE POWER SUPPLIES MUST BE LOCATED OUTSIDE THE EXAM ROOM TO RAMP DOWN THE MAGNET.**
2. **MAKE SURE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON, OR THE HATCH IS OPENED IF A MOBILE VAN, BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. RE VIEW AND FOLLOW CRYOGEN SAFETY MEASURES CONTAINED IN SECTION 5–3 OF THE INTRODUCTION (CRYOGEN SAFETY).**
3. **MAKE SURE THAT THE MAGNET IS AT LEAST <u>90%</u> FULL OF LIQUID HELIUM TO PRE VENT THE LIQUID HELIUM LEVEL FROM DROPPING TO A POINT, DURING RAMPING, WHERE A QUENCH MAY OCCUR.**
4. **METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD.**
5. **A SUPERCONDUCTING MAGNET IS AN ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY DURING A QUENCH AND CREATING A VOLTAGE OF 100V OR MORE ACROSS THE MAIN LEADS AND EXTENSIONS.**
6. **MAKE SURE INPUT POWER TO THE MAIN POWER SUPPLY IS DISCONNECTED WHEN CONNECTING MAIN POWER LEADS AND THE POSITIVE AND NEGATIVE POWER LUGS DO NOT CONNECT TO EACH OTHER.**



RAMPDOWN FLOWCHART
ILLUSTRATION 2–1

2–1 PREPARATION FOR RAMP DOWN

Note

The Rampdown procedure requires approximately 5 hours to complete.

1. “Top” fill the magnet to at least 90% helium level before continuing with this procedure.
2. Make sure Shim Lead Assembly is “Engaged” in conformance with SET UP AND CALIBRATION, Section 6. If the Shim Lead Assembly cannot be “Engaged”, the magnet can be ramped down with the Auxiliary Rampdown Cable as shown in SCHEMATICS/INTERCONNECTS, Illustration 2–1.
3. Set up the field monitoring equipment Probe and Teslameter in conformance with SET UP AND CALIBRATION, Section 5.
4. Perform Ramp Power Supply checks in conformance with SET UP AND CALIBRATION, Section 7. Verify Shim Supply Heater currents are set at 810 mA $\pm$ 10 mA.
5. Make sure that the Input Power Cables for the Power Supplies are disconnected.
6. Replace the Ramp Lead Contact Bands in conformance to REPLACEMENT/MAINTENANCE, Section 13.
7. Connect the Shim Power Supply to the magnet by making all cable connections In conformance with SET UP AND CALIBRATION, Section 8.
8. Remove the Shim Lead Vent Cap and allow frost to appear on the Shim Lead Connector Housing before removing shim currents.
9. Connect the input power supply cable to the Shim Power Supply.

Note

This section allows one to remove all shim currents to find the actual main field to document and use as a target parking frequency when re–ramping the magnet. This will enable parking the magnet at a more accurate frequency and prevent retuning the rest of the system.

WARNING!

BEFORE CONTINUING THE RAMP DOWN PROCEDURE, THE MAGNET CRYOGEN LEVEL SHOULD BE NO LESS THAN 90% TO PREVENT A POSSIBLE QUENCH.

MAKE SURE THE SHIM LEAD VENT CAP IS REMOVED AND FROSTING IS VISIBLE ON THE SHIM LEAD CONNECTOR BOX PRIOR TO TURNING ON THE SHIM POWER SUPPLY.

10. Set the MAIN POWER Switch (Shim Power Supply) to the ON position.
11. Dial in all last recorded Transverse 1 Currents from DATA SHEETS, Section 8 into the Shim Supply. Make sure the Current polarities are correct.
12. Set Transverse 1 Heater Switch (Shim Power Supply) to 1 (on). Verify that the heater current is 810mA $\pm$10mA. If it is not correct, adjust it with the adjustment screw located in the rear of the Shim Power Supply. Allow a few minutes for the heater to drive the switches resistive.

2–1 PREPARATION FOR RAMP DOWN (continued)

Note
The field strength reading on the Teslameter should change as the Axial Shim currents are adjusted down to zero. If no change in the Teslameter reading is noticed, the Axial Heater Switch may still be “persistent; if this occurs a delay of about 2 more minutes is necessary before continuing to decrease currents.

13. Slowly adjust all Shim CURRENT controls to zero.
14. Set Heater Switch to 0 (off).
15. Repeat Steps 11 through 14 for Transverse 2 and Axial Shim Coils.
16. Set the MAIN POWER Switch (Shim Power Supply) to OFF and disconnect all cables.

Note
The frequency displayed on the Teslameter after the shim currents are removed is the target frequency for when the magnet is reramped.

17. Record the frequency displayed on the Teslameter in the DATA SHEET Tab, Table 6–1.
18. Replace the Shim Lead Vent Cap.

Note
All “Main Coil Driving Voltages” provided in this procedure will be equal in magnitude and opposite in polarity for “Reversed Ramped” magnets.

2–2 RESISTANCE CHECKS

1. Make sure that Input Power Cable for the Magnet Power Supply is disconnected.
2. Connect the Magnet Power Supply and Main Lead Extensions to the magnet by making all cable connections in conformance with SET UP AND CALIBRATION, Section 8. (“Electrical Connections For Ramping And Shimming”)
3. Set HEATER 1 MAIN and HEATER 2 SHIM AXIAL switches to the 0 (off) position.
4. Set CURRENT ADJUST and VOLTAGE controls to 0 (full CCW).
5. Connect the Input Power Cable for the Magnet Power Supply.

WARNING!

MAKE SURE MAIN HEATER SWITCH IS OFF DURING THE RESISTANCE CHECKS. SETTING THE MAIN HEATER SWITCH TO O (ON), DURING THE RESISTANCE CHECKS, WILL CAUSE A MAGNET QUENCH.

6. Set the MAIN POWER and POWER ON switches to ON.

2–2 RESISTANCE CHECKS (continued)

7. Set HEATER 2 SHIM AXIAL Switch to 1 (on) and observe current rise in Analog Ammeter (800–820 mA) to verify circuit continuity. Make sure Main Heater Switch is off.
8. Connect a Digital Voltmeter (DVM) to the end of the Voltage Sense Leads.
9. Set CURRENT ADJUST controls on Magnet Power Supply to maximum (full CW).
10. Observe the Digital CURRENT Meter and slowly turn the VOLTAGE control (CW) to set 500A current through the Main Power Leads, Lead Extensions and persistent Main Switch.
11. Record the voltage reading on the (DVM) in DATA SHEET Tab, Table 6–1.

WARNING!

A VOLTAGE READING GREATER THAN 150 MILLIVOLTS AT 500 AMPS INDICATES UNACCEPTABLE INTERNAL CONTACT RESISTANCE OF THE LEAD EXTENSIONS. HIGHER RESISTANCES WILL ADD MORE HEAT TO THE MAGNET INCREASING BOILOFF AND POSSIBLY CAUSING A QUENCH DURING RAMPING.

12. Perform one or more of the bulleted steps below, as necessary, if the DVM voltage is greater than 150mV.
    - Wait approximately 1 minute with the current running, readings may drop as the Power Lead Extensions cool.
    - Tighten the nuts on top of the Hold Down Tool.
    - If the reading still exceeds 150 mV: turn the VOLTAGE and CURRENT ADJUST controls to zero (full CCW), turn off Magnet Power Supply input power, then check/tighten the bolts securing the Ramp Cables to the Power Supply and Ramp Leads Extensions. Lift and reseat the Ramp Leads. Repeat Steps 6 through 8.
13. Continue to Step 15 after the DVM voltage is less than 150mV.
14. Gradually increase the VOLTAGE ADJUST control to pass 735A through the Main Power Leads, Lead Extensions and persistent Main Switch while observing the Power Supply Voltmeter. If the voltage exceeds 2.2V, discontinue the test and perform Step 12 above.
15. Set the power supply VOLTMETER SELECT SWITCH to MAIN POWER SUPPLY position (This will display the output of the power supply monitored at the output lugs). A voltage less than 2.2V at 735A indicates acceptable system resistance. If the voltage exceeds 2.2V during the test, follow the procedures in Step 12 for adjusting contact resistance.
16. Turn the CURRENT ADJUST and VOLTAGE controls off (full CCW) and continue with Section 2–3, MAGNET RAMPDOWN.

2–3 MAGNET RAMPDOWN (DECREASE TO ZERO)

WARNING!

THE SIV MAGNET CAN BE QUENCHED IF THE MAGNET POWER SUPPLY EXPERIENCES LARGE OUTPUT VOLTAGE FLUCTUATIONS AND/OR EXCESSIVE RIPPLE. MAKE SURE THE POWER SUPPLY IS CHECKED FOR THE ABOVE AS REQUIRED DURING CALIBRATION. RIPPLE CAN ONLY BE ACCURATELY CHECKED AND ADJUSTED BY THE VENDOR.

WARNING!

AXIAL SHIM SWITCH HEATERS <u>MUST</u> REMAIN ON DURING THE ENTIRE RAMP DOWN PROCESS TO PREVENT IRREPARABLE SHIM COIL DAMAGE AND MAGNET QUENCH DURING RAMPING. THE POWER SUPPLY WILL NOT PASS CURRENT IN THE MAIN POWER LEAD CIRCUIT WITH THE AXIAL SHIM HEATERS OFF.

CAUTION

If a Quench occurs during change of magnetic field, immediately turn VOLTAGE control and CURRENT control to zero.

2–3 MAGNET RAMPDOWN (DECREASE TO ZERO) (continued)

WARNING!

MAKE SURE THE POLARITY RECORDED IN THE DATA SHEET TAB MATCHES THE CONFIGURATION OF THE RAMP POLARITY PLATE ON THE SHIM LEAD ASSEMBLY. THE "NORMAL" RAMP POLARITY IS "+" ON THE LEFT AND "–" ON THE RIGHT AS VIEWED FROM THE COLD HEAD SIDE OF THE MAGNET. "REVERSED" RAMP POLARITY HAS "+" ON THE RIGHT AND "–" ON THE LEFT AS VIEWED FROM THE COLD HEAD SIDE OF THE MAGNET. IT THE TWO DO NOT MATCH, CALL THE NATIONAL SUPPORT CENTER TO AVOID A QUENCH.

WARNING!

MAKE SURE THAT THE CONNECTION POLARITY AND POWER SUPPLY CURRENT ARE THE SAME AS THE LAST RECORD IN TABLE 6–1 OF DATA SHEETS. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH.

Note

Ice will build up around the Ramp Lead Hold Down Tool Flow Holes during ramping down. Remove ice as needed to maintain helium gas flow through the Flow holes.

1. Retrieve the Main Coil Connection Polarity and latest final ramping current in the DATA SHEET Tab, Table 6–1.

Note

The Axial Shim Heater must remain on throughout the Rampdown procedure.

2. Make sure the Axial Shim Heater is on and Ramp Leads are connected with the polarity indicated in step 1.
3. Set the power supply VOLTMETER SELECT SWITCH to MAIN COIL position.
4. Set CURRENT ADJUST controls on power supply to maximum (full clockwise).
5. Set VOLTAGE control to adjust power supply output current to the Parking Current value obtained in Step 1 above.
6. Set the HEATER 1 MAIN Switch to 1 (on).
7. Allow approximately 1 minute for the Main Switch to go normal.
8. When the Main Switch is normal, <u>slowly</u> decrease VOLTAGE control, counterclockwise, until negative 0.3 volts (–0.3 V) is observed across the Power Supply Voltmeter (Main Coil Position).

2–3 MAGNET RAMPDOWN (DECREASE TO ZERO) (continued)

9. When the main field decreases to 1.3T, slowly adjust the VOLTAGE control to negative 1.5 volts across the Main Coil. Carefully adjust the VOLTAGE control to maintain negative 1.5 volts across the Main Coil until the field is ramped down to 1.0T (10,000 gauss).
10. When the field decreases to 1.0T, slowly adjust the VOLTAGE control to minimum (full counterclockwise).
11. After performing Step 10, adjust the CURRENT ADJUST controls to minimum (full counterclockwise).
12. Observe voltmeter reading with power supply VOLTMETER SELECT SWITCH in the MAIN COIL position. A zero reading, on the power supply Digital VOLTAGE and CURRENT meters, indicates that the magnet is fully discharged.
13. When the magnet is full discharged, set the HEATER 1 MAIN and THE HEATER 2 SHIM AXIAL Switches to 0 (off).
14. Set the MAIN POWER and POWER ON Switches to OFF.
15. Disconnect input power cable from Magnet Power Supply.

2–3 MAGNET RAMPDOWN (DECREASE TO ZERO) (continued)

CAUTION

Replace Ramp Port Caps immediately after removing Main Power Lead Extensions to prevent ice build up inside Vertical Stack.

16. Open Vent Valve (V2) to de–pressurize the Cryostat to 0.25 psig. Close V2.
17. Disconnect the Main Power Leads on the top of the magnet and remove the Main Power Lead Extensions. Immediately replace caps on Main Power Lead Extension Receptacles.

CAUTION

Cryostat exhaust flow rates and pressure must be checked and adjusted as required after magnet installation, ramping and shimming to ensure that proper cooling conditions are maintained and no leaks are present in the Helium Exhaust System or Vent Valve (V2).

Note
Read all flow rates from the bottom of the float (ball) on the flow meters.

Note
Flow rates may be temporarily elevated after ramping. Do not adjust them until after the magnet has had time to stabilize (at least one day).

18. Make sure the following conditions are maintained. Re–check settings in three days and again after one week:

INSTRUMENTATION LEAD FLOWMETER (F1) = 0.8 – 1.2 SCFH
SHIM LEAD FLOWMETER (F2) = 1.8 – 2.2 SCFH
CRYOSTAT GAUGE PRESSURE = 0.25 – 0.50 psig

Note
If flow rate through F2 is less than 1.8 SCFH or the pressure gauge reads less than 0.25 psig, pressurize the vessel and “bubble test” all exhaust plumbing joints, relief valve and Shim Lead Connector. Make sure V2 is fully closed. Repair any leaks.